

■ Project Spotlight: Zenith Retail Annual Sales Report 2024 (Excel)

I recently completed an end-to-end sales analysis project for a fictional retail chain, "Zenith Retail," using Microsoft Excel — from messy raw data to a fully interactive dashboard. Sharing the journey below ■

■ *Project Objective*

Zenith Retail had a year's worth of transactional sales data (2,000+ orders across regions, stores, and product categories) sitting in a raw, unstructured format. The goal was to clean it, structure it, and build a dashboard that leadership could use to quickly answer questions like:

- Which regions and stores are driving revenue?
- What's selling, and how much?
- How are customers paying?
- How do sales trend month over month?

■ *Data Cleaning Challenges Solved*

Real-world data is rarely clean, and this dataset was intentionally built to prove it. Some of the issues I identified and fixed:

- Order dates stored as text in 5 different formats (YYYY-MM-DD, DD/MM/YYYY, DD-Mon-YYYY, etc.) — standardized into a single format
- Inconsistent casing/spacing in Region and Product Category fields (e.g., "north", "NORTH", " North") — cleaned and normalized
- Unit Price stored as text with "Rs." prefixes in some rows — converted to usable numeric values
- Discount column mixing decimals (0.10) and whole numbers (10) meant to represent the same 10% — standardized
- Duplicate Order IDs, ~20 fully duplicated rows, and ~15 blank rows scattered through the sheet — identified and removed
- Blank and negative Quantity values (likely returns) — investigated and handled with a clear treatment rule

Only after this cleanup did I build out pivot tables and pivot charts on top of the data — because a dashboard is only as trustworthy as the data behind it.

■ *Key Findings*

- North (22%) and West (21%) are the top-performing regions; Central lags at 16%
- Chandigarh Store led all locations in revenue (■10.9M), followed by Mumbai and Nagpur
- Net Banking (18%) was the most-used payment method, but usage was fairly even across Cash, Credit Card, Debit Card, EMI, and UPI — no single method dominates
- Home Appliances and Electronics were the top two product categories by quantity sold
- December and January were peak sales months, while February dipped to the year's lowest — useful signal for planning inventory and staffing around seasonality

- Order volume stayed relatively steady (150–200/month) even as revenue fluctuated, suggesting average order value — not just order count — is what's driving the swings

■ ■ *How It Was Built*

- Cleaned and structured raw data using Excel formulas and data validation
- Built 5+ pivot tables to summarize sales by region, store, category, and payment method
- Created pivot charts (bar, line, donut, pie) for visual storytelling
- Designed a single-page interactive dashboard with slicers for Region, Month, and Product Category — allowing dynamic filtering without touching the underlying data
- Highlighted a "Top 5 Stores" view to make performance rankings instantly visible

■ *Takeaway*

This project reinforced something I keep relearning as a data analyst: the unglamorous work of cleaning and validating data is what makes the "pretty" dashboard actually useful. Anyone can make a chart — the real value is in trusting the numbers behind it.

Would love to hear how others would approach this dataset — if you were in my place, how would you have handled the cleaning or the dashboard differently? Always looking to sharpen my process!

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